

Supplementary Material for Static Feasibility Can Overstate Chronological Admissibility in Distributional Power-System Operating States

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I. SCOPE AND PROVENANCE

This supplement provides computational details for the chronology, alternating-current (AC) restoration, Great Britain, Elexon operational-data, and independent numerical-verification analyses reported in the main article. The RTS calculations use the same admitted operating-law states and source-processing definitions throughout the study, with RTS-GMLC pinned to source commit 3ece0d3725c844056132393ee252b3083dd4eab4 [1], [2]. The chronology and AC calculations add constraints and validation layers while preserving the static mean, shape, hour-conditioned, and fixed-commitment state definitions. We use direct-current (DC) network constraints for chronological dispatch, mixed-integer linear programs (MILPs) for commitment, linear programs (LPs) for continuous relaxations, and nonlinear programs (NLPs) for AC projection.

II. NATIVE CHRONOLOGY FORMULATION

Let g index the 41 admitted generation coordinates and $t = 0, \dots, T-1$ the hourly periods, with $T = 168$. Active generation $p_{g,t}$ is measured in MW. For thermal units, $P_g^{\min}$ is minimum online output, $P_{g,t}^{\max}$ is hourly available maximum output, $u_{g,t} \in \{0, 1\}$ is on/off status, and $y_{g,t}, z_{g,t} \in \{0, 1\}$ are startup/shutdown indicators. The independent verification formulation imposes the power/status link at every hour and the transition equations only for $t = 1, \dots, T-1$:

$$P_g^{\min} u_{g,t} \leq p_{g,t} \leq P_{g,t}^{\max} u_{g,t}, \quad (1)$$

$$u_{g,t} - u_{g,t-1} = y_{g,t} - z_{g,t}, \quad (2)$$

$$y_{g,t} + z_{g,t} \leq 1, \quad (3)$$

For source minimum-up time U_g and minimum-down time D_g in hours, observed starts and stops satisfy, for $t = 1, \dots, T-1$,

$$\sum_{\tau=\max(1,t-U_g+1)}^t y_{g,\tau} \leq u_{g,t}, \quad (4)$$

$$\sum_{\tau=\max(1,t-D_g+1)}^t z_{g,\tau} \leq 1 - u_{g,t}. \quad (5)$$

The integers U_g and D_g are the source Min Up Time Hr and Min Down Time Hr fields (rounded upward only if a noninteger were encountered). Hourly active-power balance is enforced; hydro availability follows the retained source-processing treatment used for the static baseline. The boundary is intentionally weak: $u_{g,0}$ is free, $y_{g,0} = z_{g,0} = 0$, and no transition to a pre-horizon status is imposed. Residence-time constraints apply only within the observed week, with no terminal extension. This makes the gate a relaxation, so exact-mean infeasibility is inherited by any stricter network-constrained chronology.

The native online-to-online ramp-only gate uses Ramp Rate MW/Min $\times 60$ for the hourly interval. Unit-commitment (UC) benchmarks can impose different ramp, startup and boundary semantics [3], [4]; the reported chronology conclusions here use the pinned native RTS parameters throughout. Table I summarizes admission under the native no-network relaxation.

III. CHRONOLOGICAL MEAN REPAIR

For a target vector of mean generator outputs μ_T , let $\mu_{T,g}$ be its coordinate for generator g and let $d_g^+, d_g^- \geq 0$ be positive and negative mean deviations in MW. The repair problem minimizes their sum, equal at optimality to the L_1 deviation, where $\|v\|_1 = \sum_g |v_g|$:

$$\sum_g (d_g^+ + d_g^-), \quad \frac{1}{T} \sum_{t=0}^{T-1} p_{g,t} - d_g^+ + d_g^- = \mu_{T,g}, \quad (6)$$

TABLE I
HARMONIZED EXACT-MEAN ADMISSION IN THE SOURCE-NATIVE NO-NETWORK RELAXATION.

Constraint family	March	July
Binary commitment only	Admitted	Admitted
Native online ramp (redundant)	Admitted	Admitted
Source-native minimum up/down	Rejected	Rejected
Ramp + minimum up/down	Rejected	Rejected
Independent y/z min-up/down check	Rejected	Rejected

TABLE II
JULY CHRONOLOGICAL MEAN-REPAIR EVIDENCE. THE EXACT RELAXATION VALUE IS ROUNDED TO NEAREST; BRACKET ENDPOINTS ARE ROUNDED OUTWARD.

Declared set	Lower MW	Upper MW	Status
No-network min-up/down relaxation	8.020508	8.020508	Exact
Network + min-up/down + native ramp	8.020508	11.926297	Certified bracket

subject to the relaxed chronology. July solves to an exact objective of 8.0205084643 MW with reported mixed-integer-program (MIP) gap zero. March does not have a certified integer incumbent within the declared solver budget; its LP relaxation yields 40.0187592756 MW, so that number is a lower bound only.

For July, the stricter Area-1 DC network chronology is also bounded directly. Let $A_{\text{chron}}^{\text{net}+\text{ramp}}(\mu_{\text{Jul}})$ denote the minimum L_1 mean deviation from the July target under nodal DC balance, all 38 internal continuous line ratings, source-native minimum up/down logic, and native online-to-online ramping. The certified interval, with endpoints rounded outward, is

$$8.0205084642 \leq A_{\text{chron}}^{\text{net}+\text{ramp}}(\mu_{\text{Jul}}) \leq 11.9262964643 \text{ MW}. \quad (7)$$

The lower endpoint is inherited from the exact no-network relaxation. The upper endpoint is not an optimality claim: it is an explicit feasible schedule obtained after a bounded network MIP search and status-neighborhood refinement. A separate verifier recomputes the schedule from the source data and records maximum system-balance residual 6.6×10^{-12} MW, maximum branch-loading fraction 1.0000000000000002, zero active-power bound violation, zero native online-ramp excess, and zero minimum-up/down violations. The exact network optimum remains unresolved inside the interval.

Table II summarizes the July repair bounds. Table III reports the largest generator-mean adjustments in the exact relaxed repair.

IV. BENCHMARK SEMANTICS

Published unit-commitment benchmarks motivate careful separation of ramp, startup and boundary conventions. Historical PGLib stress calculations are not part of the candidate's numerical evidence because their exact input revision has not been established. The native RTS results and the independent residence test are separated from that comparison.

V. DIRECT AC FEASIBILITY PROJECTION

The AC projection uses the Area-1 internal 24-bus/38-branch topology. At bus i , let V_i be voltage magnitude in per-unit (p.u.), θ_i its phase angle in radians, and $v_i = V_i e^{j\theta_i}$ its complex voltage phasor, with $j^2 = -1$. Let Y_{ik} be an entry of the per-unit bus-admittance matrix, $\mathcal{G}_i$ the set of generators at bus i , P_g, Q_g generator active/reactive power, and P_i^d, Q_i^d active/reactive demand. With active power in MW, reactive power in MVar, and base power $S_{\text{base}} = 100$ MVA, the complex nodal balance is

$$\sum_{g \in \mathcal{G}_i} (P_g + jQ_g) - (P_i^d + jQ_i^d) = S_{\text{base}} v_i \overline{\sum_k Y_{ik} v_k}. \quad (8)$$

The overbar denotes complex conjugation and k indexes buses. For the DC target generator output P_g^{DC} , the restoration objective is $\sum_g |P_g - P_g^{\text{DC}}|$ in MW. The nonlinear program imposes source hourly P availability, source generator Q limits, $0.95 \leq V_i \leq 1.05$ p.u., and apparent-power limits at both branch terminals: $|S_{ik}| \leq S_{ik}^{\text{cont}}$ and $|S_{ki}| \leq S_{ki}^{\text{cont}}$. Here S_{ik} is complex power injected into the branch at terminal i , and S_{ik}^{cont} is the continuous MVA rating. The implementation adds the numerical voltage regularizer $10^{-8} \sum_i (V_i - 1)^2$, whose value is at most 6×10^{-10} over the declared voltage interval and 24 buses. Reported restoration costs exclude that term. Rooftop photovoltaic generation is fixed negative active load, and bus reactive demand is scaled by the same regional load factor used for active demand. CasADi and IPOPT provide the primary witness search [5], [6]; successful points are checked against all constraint residuals. For hour t , $\hat{R}_t^{\text{AC}}$ denotes the returned L_1 witness cost, which upper-bounds the ideal minimum R_t^{AC} for a feasible returned point.

TABLE III

LARGEST COORDINATE CHANGES IN THE EXACT JULY REPAIR FOR THE NO-NETWORK MINIMUM-UP/DOWN RELAXATION. ABSOLUTE CHANGES ARE BETWEEN REPAIRED AND TARGET GENERATOR MEANS.

Generator	Target mean MW	Repaired mean MW	Abs. change MW
122_WIND_1	65.145	63.484	1.662
118_CC_1	47.385	48.347	0.961
102_CT_2	0.000	0.892	0.892
102_CT_1	0.000	0.810	0.810
102_STEAM_4	72.381	71.655	0.726
115_STEAM_3	151.012	150.571	0.440
101_CT_1	0.000	0.420	0.420
113_PV_1	26.705	26.290	0.415
123_STEAM_2	154.779	154.365	0.414
101_CT_2	0.000	0.395	0.395

TABLE IV

PRIMARY SHARED-CONTINUOUS AC PROJECTION. MAXIMUM CONSTRAINT VIOLATION RETAINS THE IMPLEMENTED CONSTRAINT SCALING IN MW, MVAR, OR MVA².

Quantity	March	July
Prespecified hours	24	24
Successful witnesses	24	24
Success rate	100%	100%
Median L_1 restoration (MW)	39.208762	31.080193
Minimum restoration (MW)	18.799818	16.923903
Maximum restoration (MW)	74.424020	68.474157
Minimum voltage (p.u.)	0.976920	0.988729
Maximum voltage (p.u.)	1.050000	1.050000
Maximum branch loading fraction	1.000000	1.000000
Maximum constraint violation	9.50×10^{-8}	8.67×10^{-8}

This direct projection differs from a heuristic recovery pipeline using distributed slack and switching between active-power/voltage-controlled (PV) buses and active/reactive-power-specified (PQ) buses. The latter is well motivated in the literature [7], but here P , Q , voltage, and branch limits are optimized jointly. The benefit is a direct feasible witness; the nonconvex program provides a locally computed restoration value rather than a globally certified minimum. Table IV summarizes the shared-continuous witness results.

The median restoration is equal to the active-loss correction to the precision shown in the released summary. This is an equality of the reported medians, not a claim of coordinatewise equality or of equality at every hour. Each returned local solution redistributes active generation subject to the declared AC constraints.

VI. FIXED-COMMITMENT RECOVERY AND SOLVER BOUNDARY

Table V reports the bounded fixed-commitment witness searches. The portable routine uses at most 250 IPOPT iterations and 0.4 CPU seconds per initial guess, with a source-data guess and a previous-hour solution when available. The fixed-commitment mode sets thermal units that are off in the target DC schedule to zero and retains minimum output for units that are on. This branch is kept as a supporting witness search, not as a comparative success-rate experiment. Under one bounded attempt, 8 March and 15 July hours returned verified witnesses. Repeating the portable bounded routine with the same mathematical model but a different wall-clock realization returned 10 March and 15 July witnesses. The nested March set gained hours 1 and 2; no previously verified witness was lost. This solver-budget sensitivity is why the manuscript does not report the bounded return count as a physical rate.

A separate PYPOWER 5.1.18 AC-OPF recovery independently converged for the first three March fixed-pattern hours. Its restoration values were 50.666395, 50.347203, and 50.107906 MW. The CasADi/IPOPT values for those same hours were 50.666127, 50.346555, and 50.107658 MW. Absolute differences are below 0.001 MW. These three common successful cases provide a cross-solver implementation check; PYPOWER or IPOPT nonconvergence on other cases is not used as a physical rejection criterion [8].

TABLE V
BOUNDED FIXED-PATTERN WITNESS SEARCHES (DIAGNOSTIC ONLY).

Attempt	March	July
Archived bounded search	8/24	15/24
Portable bounded rerun	10/24	15/24

VII. PER-HOUR SHARED AC RESULTS

Table VI reports every shared-continuous hourly witness. For each hour, $V_{\min}$ is the minimum bus-voltage magnitude, and the maximum loading fraction is the largest terminal apparent power divided by its continuous branch rating.

TABLE VI: Shared-continuous AC projection by hour. Month 3/7 denotes March/July; hour 00–23 is the hour of the first target day. Voltage is in p.u.; loading fraction is dimensionless.

Month	Hour	$\hat{R}_t^{AC}$ MW	Loss MW	$V_{\min}$	Max load frac.
3	00	39.347959	39.347959	0.991926	0.811155
3	01	39.069565	39.069565	0.991901	0.810999
3	02	38.861550	38.861550	0.991882	0.810884
3	03	35.917636	35.917636	0.991477	0.804338
3	04	43.296760	43.296760	0.992287	0.813389
3	05	40.743685	40.743685	0.992725	0.818524
3	06	35.653200	35.653200	0.992155	0.799467
3	07	18.799818	18.799818	0.992516	0.797875
3	08	42.137514	42.137514	0.993159	0.883214
3	09	50.658318	50.658318	0.992880	0.970142
3	10	45.834138	45.834138	0.992954	0.910159
3	11	27.512959	27.512959	0.992775	0.800609
3	12	26.095387	26.095387	0.992775	0.800499
3	13	26.560962	26.560962	0.992891	0.801572
3	14	20.113626	20.113626	0.992536	0.799363
3	15	19.000586	19.000586	0.992482	0.797941
3	16	36.167438	36.167438	0.993260	0.811372
3	17	26.937829	26.937829	0.992384	0.797816
3	18	65.129733	65.129733	1.002656	0.857504
3	19	74.424020	74.424020	1.001644	0.931352
3	20	65.983801	65.983801	0.998535	0.883202
3	21	53.946111	53.946111	0.993830	0.821004
3	22	56.940689	56.940689	0.978812	1.000000
3	23	56.830450	56.830450	0.976920	1.000000
7	00	28.327757	28.327757	0.995792	0.814364
7	01	30.758927	30.758927	0.997287	0.810216
7	02	36.081598	36.081598	0.996748	0.802268
7	03	67.983109	67.983109	1.005548	0.873842
7	04	64.932554	64.932554	1.005083	0.852886
7	05	61.464068	61.464068	1.006624	0.841047
7	06	68.474157	68.474157	1.005813	0.897325
7	07	51.079191	51.079191	0.999561	0.820810
7	08	36.743528	36.743528	0.994198	0.819414
7	09	34.517953	34.517953	0.991242	0.832830
7	10	34.907402	34.907402	0.988729	0.826097
7	11	25.069167	25.069167	1.013893	0.865932
7	12	25.541388	25.541388	1.013462	0.852520
7	13	26.234493	26.234493	1.014577	0.853372
7	14	31.252582	31.252582	1.013600	0.859318
7	15	30.907804	30.907804	1.013694	0.943528
7	16	32.446634	32.415957	1.012533	1.000000
7	17	32.836340	32.808634	1.014336	1.000000
7	18	30.328717	30.304649	1.015278	1.000000
7	19	27.394640	27.372354	1.014946	1.000000
7	20	23.666486	23.666486	1.009548	0.946155
7	21	21.203827	21.203827	1.007806	0.848671
7	22	16.923903	16.923903	1.001042	0.832410
7	23	27.359009	27.359009	0.993264	0.826480

VIII. INDEPENDENT PyPSA-GB REPLICATION

The independent-system extension uses PyPSA-GB [9] at pinned source commit

8e084afe4fb2d4be86f270d3f12ad3315eee2a3a.

TABLE VII
PYPSA-GB JANUARY–JULY INTERNAL-GENERATION REPLICATION.

Quantity	Value
Common internal coordinates	2685
Mean movement M (MW)	16785.255
Full empirical W_1 (MW)	18651.979
Shape gap (MW)	1866.724
Shape share	10.01%
Clock-hour W_1 (MW)	19888.552
Clock-hour premium (MW)	1236.573
Clock-hour premium / full W_1	6.63%
Load shedding, Jan./Jul. (MWh)	0 / 0

Two Historical-2020 Reduced-network scenarios cover Jan. 1–7 and Jul. 1–7 at 60-min resolution. Both are solved with HiGHS in LP mode and terminate optimally with zero load shedding. The returned solved-network diagnostics report 35 buses, 99 lines, 7 links, and 74 storage units in each week. The weather cutout `uk-2020.nc` is verified at 766,913,074 bytes with MD5 9d08ab99530454944a8efffcd547b720.

The primary operating-law state is defined on the 2685 generator coordinates common to both weeks after removing the `EU_import` carrier. This exclusion is deliberate because the historical PyPSA-GB workflow represents fixed interconnector supply as generator injections; storage units and links are also kept outside the state vector. Write M for the L_1 difference between the source and target mean vectors, W_1 for the first Wasserstein distance between the empirical operating laws with full-coordinate L_1 ground cost, and $G_{\text{shape}} = W_1 - M$ for the shape gap. On that primary internal-generation universe,

$$M = 16785.255046 \text{ MW}, \quad W_1 = 18651.979268 \text{ MW},$$

so

$$G_{\text{shape}} = 1866.724221 \text{ MW}, \quad \frac{G_{\text{shape}}}{W_1} = 10.0082\%.$$

When transport assignments are restricted to source and target hours with the same clock hour, the empirical transport rises to 19888.552386 MW. The conditioning premium over unrestricted full W_1 is therefore 1236.573118 MW, or 6.6297%.

Table VII summarizes the internal-generation comparison; each percentage uses full empirical W_1 as its denominator.

Twenty-one internal generator coordinates appear in July but not January. In the primary comparison they are excluded to keep the state dimension fixed. A union-universe sensitivity zero-pads those coordinates in January. This changes M from 16785.255046 to 16797.652710 MW and full W_1 from 18651.979268 to 18664.376932 MW, while leaving G_{shape} equal to 1866.724221 MW to numerical precision. The July-only coordinates contribute 12.397664 MW to the target mean. Thus the shape result is not an artifact of the common-coordinate intersection.

The geocoding failure-cache text in `nominatim_cache_failures.txt` was normalized from CP1252 to UTF-8 without changing capacity, topology, time-series, or optimization inputs. The run manifest records before/after SHA-256 values for that compatibility change.

This GB extension replicates the distributional diagnostic on an independently developed model. Its LP extracts alone do not establish absence of chronology parameters upstream. The pinned source contains a thermal-fuel route with unresolved unit/carrier-mapping issues and an explicit wholesale overlay for combined-cycle gas turbines (CCGT), open-cycle gas turbines (OCGT), coal and oil. The overlay parameters are source-defined optional assumptions; the original network experiment remains unevaluated because the full time-dependent network and eligibility flags were not archived. The detailed parameter-admission table records each source, unit, code path and limitation.

IX. ELEXON OPERATIONAL-DATA EVIDENCE

The record identifiers refer to balancing mechanism units (BMUs), the Dispute Final (DF) settlement run, and Physical Notifications (PNs). Dynamic-parameter codes identify Minimum Non-Zero Time (MNZT), Minimum Zero Time (MZT), Run Up Rate Export (RURE), and Run Down Rate Export (RDRE). The operational-data extension uses public Elexon Insights API records [10]. The comparison dates are Wednesday Jan. 11 and Wednesday Jul. 12, 2023. B1610 quantities are settlement-period metered energy in MWh; the released analysis converts each 30-min quantity to average MW by multiplying by two. Both raw files contain 48 settlement periods, no duplicate BMU/period rows in the selected state, and only the DF settlement-run type.

The acquisition-date BMU reference table identifies transmission-connected production BMUs. Of these, 260 have complete B1610 records on both days. Equal-weight L_1 transport on the resulting 48×260 states gives

$$M = 19553.647583 \text{ MW}, \quad W_1 = 20776.837250 \text{ MW},$$

and hence

$$G_{\text{shape}} = 1223.189667 \text{ MW}, \quad G_{\text{shape}}/W_1 = 5.8873\%.$$

Restricting assignments to the same settlement-period number gives 21016.487667 MW, a 239.650417-MW (1.15345%) premium. In the 268-coordinate zero-padded union, both M and W_1 increase by 2.081792 MW and the shape gap is unchanged to numerical precision.

Physical Notification segments are integrated piecewise linearly over complete 1800-s settlement periods. For 249 complete January coordinates, the mean period-level L1 PN-to-B1610 discrepancy is 3461.403181 MW; for 252 July coordinates it is 3141.137250 MW. These are descriptive differences, not inferred balancing volumes, feasibility repairs, or causal effects. A deterministic twelve-BMU sample additionally freezes effective MNZT, MZT, RURE, and RDRE snapshots. It demonstrates that nonzero temporal parameters exist in the public data model but is not used as a system-wide chronology calibration.

The local acquisition records retain raw responses with SHA-256 hashes. The public baseline bundle contains acquisition code and derived tables, but excludes raw third-party responses; the new candidate adds the sensitivity acquisition register and derived results. The registry was acquired on Sep. 20, 2026 and is not asserted to reproduce historical 2023 registration state.

X. NETWORK REPAIR BOUNDS AND INDEPENDENT NUMERICAL VERIFICATION

A warm-started HiGHS 1.12.0 run re-tested the July network/minimum-up/down/native-ramp MIP for 600 s. The verified 11.9262964643-MW witness was accepted with zero bound and integrality violation. At the time limit, the primal objective was 11.926296464290 MW, the dual bound was 8.020508464288 MW, and the maximum primal infeasibility was 7.951×10^{-12} . The run therefore leaves the published bracket unchanged and does not convert the upper endpoint into an optimum.

Separately, an independent arithmetic script imports none of the primary analysis modules. It reads released CSV/ZIP artifacts, reconstructs L_1 assignment, and rechecks all seven RTS mean/full- W_1 pairs, six GB quantities, two chronological repair sums, and four AC summary claims. All 26 claims pass their declared tolerances. This validates the mapping from released artifacts to reported numbers; it does not replace the solver-status evidence used for exactness, lower bounds, or feasible witnesses.

XI. EIGHT-WEEK CHRONOLOGY AND INDEPENDENT EXPLANATION

The fixed seasonal protocol includes all eight archived weeks: January, February, March, April, May, June, July and October. The same 41 target coordinate means and native hourly inputs are used. All eight no-network minimum-up/down models terminate Infeasible with HiGHS 1.12.0, a 60-s limit, one thread and seed zero. Every archived static schedule itself passes native online-to-online ramps. For every thermal unit, let r_g be its native ramp rate in MW/min and $P_g^{\max}$ its constant available maximum. Then $60r_g \geq P_g^{\max} - P_g^{\min}$, so this on/on-only family is redundant at hourly resolution; the smallest slack is 30 MW. This does not address startup/shutdown limits or a finer time grid. A May ramp search timed out without an incumbent, but its original schedule is a directly verified witness. Both records are retained.

An exploratory independent check explains the rejection without an optimization solver. For thermal unit g with $P_g^{\min} > 0$, let $\Delta t = 1$ h and $E_g = \Delta t \sum_t p_{g,t}^{\text{target}}$ MWh. Propagate necessary lower and upper hourly generation bounds using aggregate balance, each coordinate's fixed energy, generator availability, fixed hydro and the thermal off-or-online-output gap. A positive necessary lower bound forces a unit on; an upper bound below its minimum online output forces it off. For each unit, a finite-state dynamic program stores the current status, remaining mandatory dwell and the reachable total online-hour counts. Both initial statuses start with no prior obligation, transitions add the native minimum dwell, and remaining terminal obligations are not extended beyond hour 167. Let $\mathcal{N}_g$ be the resulting reachable count set. Feasibility requires

$$\mathcal{N}_g \cap \left[\left\lceil \frac{E_g}{P_g^{\max} \Delta t} \right\rceil, \left\lfloor \frac{E_g}{P_g^{\min} \Delta t} \right\rfloor \right] \neq \emptyset.$$

The implementation widens power bounds by 10^{-5} MW and coordinate energy by 168×10^{-5} MWh; it never infers feasibility from this necessary condition. Exhaustive enumeration of all binary schedules verifies the recurrence on 1125 short-horizon forced-status cases.

Seven weeks fail this independent test. For example, January's 123_STEAM_3 target energy is 3228.521078 MWh, allowing at most 23 online hours at its 140-MW minimum output; the forced statuses and native 24-h up/48-h down times require at least 100 online hours. In July, 115_STEAM_3 is forced on at hour 2, off at hour 3 and on at hour 7, violating its eight-hour minimum downtime. June is not rejected by this single-unit test; its full mixed-integer result remains the evidence for rejection. The January source-week rejection also means that no chronologically feasible initial operating law is assumed.

XII. FIXED EIGHT-DAY ELEXON SENSITIVITY

The protocol selects every Wednesday of January and July 2023 before downloading additional dates: January 4, 11, 18, 25 and July 5, 12, 19, 26. Its four ordinal pairs are primary and its 16 cross-pairs reuse days; they are descriptive comparisons, not independent samples for significance testing. Each vector retains signed B1610 average-MW values. The acquisition-date production classification includes pumped-storage-coded units and station-demand names; no negative value is clipped. A duplicated registry identifier with consistent classification is represented once as a coordinate, with its two Energy Identification Code (EIC) aliases retained in diagnostics.

TABLE VIII
EVIDENCE SUPPORTING THE CENTRAL CLAIMS AND THE LIMITS OF EACH INFERENCE.

Claim	Evidence	Boundary
Static shared mean target admitted	Retained LP witness and analytic equality	Declared convex support only
Native hourly ramp control admits all eight weeks	Analytic redundancy plus direct archived-schedule checks	Not evidence about binding or startup/shutdown ramps
Native min-up/down rejects eight weekly means	No-network mixed-integer solves; independent necessary-condition rejection for seven weeks	Rejection transfers to stricter network chronology; repair values do not
July repair	Exact no-network relaxation plus verified network/native-ramp feasible schedule	8.020508-MW exact relaxed value (rounded); outward-rounded network repair bracket
March 40.018759-MW repair statement	LP relaxation only	8.020508–11.926297 MW
Shared AC restoration	48/48 successful direct AC NLP witnesses	Lower bound, not exact integer repair
Fixed AC supporting witnesses	Common PYPOWER/CasADi witnesses plus bounded-search records	Local nonconvex projection; no single-contingency or trajectory claim
GB replication	PyPSA-GB Jan./Jul. 2020 Reduced-network LP: nonzero shape gap and clock-hour premium, zero load shedding	Counts are solver-budget dependent; nonreturns are not infeasibility
Ellexon operational evidence	Raw B1610/PN/dynamic responses, exact hashes, and derived transport tables	Distributional diagnostic only; no GB min-up/down chronology claim
Clean-room reproduction	26/26 central numerical claims independently recomputed	Matched-day observational evidence; not a field experiment or causal redispatch estimate
		Arithmetic artifact audit; solver certificates remain separate

Pairwise complete intersections have 259–260 coordinates; the common intersection of all days has 259. Primary shape shares are 6.4889%, 5.8873%, 6.5885% and 13.5601% in ordinal order; same-period premiums range from 1.15345% to 2.46693%. Over all 16 cross-pairs, shape shares range from 3.46077% to 13.56012% and same-period premiums from 0.74634% to 9.35493%. The fixed-coordinate sensitivity gives primary shape shares of 5.89130–13.56628%. All 32 independent transportation-program checks agree with assignment results within 1.1×10^{-11} MW. Original-date quantities, run types and keys are unchanged on reacquisition despite different raw byte hashes. Numerical reproducibility is asserted within declared tolerances; byte identity requires the locked serialization environment. Two months and one weekday cannot establish annual, causal or weather-adjusted robustness.

XIII. EVIDENCE AND CLAIM LEDGER

Table VIII links each central claim to its evidence and scope.

XIV. REPRODUCIBILITY NOTES

Chronology uses SciPy’s mixed-integer interface with HiGHS [11], [12]. The reproducibility materials include the exact no-network July repair, network witness, warm-started MIP log, independent verifier, clean-room audit, PyPSA-GB parameter gate, and SHA-256 records. They also include the AC projection code, full result CSV files, per-generator correction records, solver logs, Ellexon acquisition and analysis code, and environment metadata. The failed diagnostic PYPOWER branch is retained so that nonlinear solver nonconvergence is not silently omitted. Public upstream data remain subject to their source licences and attribution requirements.

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